

Learning Activity Sheet for General Science

Quarter 3

Lesson

6

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Quarter 3: Lesson 6

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	3
Lesson No.:	6	Date:	
Lesson Title/ Topic:	Nervous and Endocrine System Integration in Thermoregulation		
Name:		Grade & Section:	

I. Activity No. 1: Survival Kit Designing for Extreme Temperature

Objective(s): By the end of the activity, the learners will be able to:

- Identify physiological challenges in extreme heat and cold.
- Explain how the nervous and endocrine systems respond to maintain homeostasis
- Design a survival kit with scientifically justified items that support body regulation.
- Collaborate and present findings using visual or digital formats

II. Materials Needed:

- -Manila paper or cartolina
- Markers, crayons, pens, sticky notes
- Scissors, tape, glue
- Printed pictures/magazines (optional)
- Digital tools (PowerPoint, Canva, Google Slides – optional)

III. Instructions:

1. Grouping and Differentiation

- Students will work in mixed-ability groups of 4-6 students, depending on the class size.
- Roles may be assigned depending on the strengths (e.g., researcher, designer, presenter)
- Students with limited access to materials may opt for digital output or simplified poster versions.
- Teacher support will be provided for students with learning difficulties through guided questions and visual aids.

2. Procedure

a. Scenario Presentation:

- *“Imagine you are preparing a survival kit for someone living in a desert or a snowy mountain. What items would you include to help the body maintain homeostasis?”*

b. Group Assignment:

- Each group selects or is assigned either a HOT (desert) or a COLD (snowy mountain) environment.

c. Discussion and Planning:

- Identify physiological challenges (e.g., dehydration, frostbite).
- Describe body responses via nervous and endocrine systems (e.g., sweating, hormone release).
- Choose survival items and explain their scientific relevance.

d. Design Phase:

- Create a poster, slide deck, or physical mock-up of the survival kit.
- Include explanations for each item:
 1. Supported body system

2. Related hormone or response
3. Role in maintaining homeostasis

e. Presentation and Comparison:

- Groups present their kits.
- Class discussion highlights differences and similarities in survival strategies.

IV. Guiding Questions:

1. Why is water more important in hot environments compared to cold ones?

2. How does clothing choice affect thermoregulation?

3. Which hormones are the most active in extreme temperatures?

4. What happens if a key survival item is missing?

5. How do the nervous and endocrine systems coordinate to maintain balance?

Rubric for the Survival Kit Design

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)
Scientific Accuracy				
Creativity and Presentation				
Team Collaboration				
Explanation of Body Responses				
Application of Concepts				

Note: This activity uses constructivist, inquiry-based, and collaborative learning strategies. It supports the IDF by integrating real-world survival scenarios, linking body systems to environmental challenges, group work and peer learning, and designing kits with visual and scientific elements